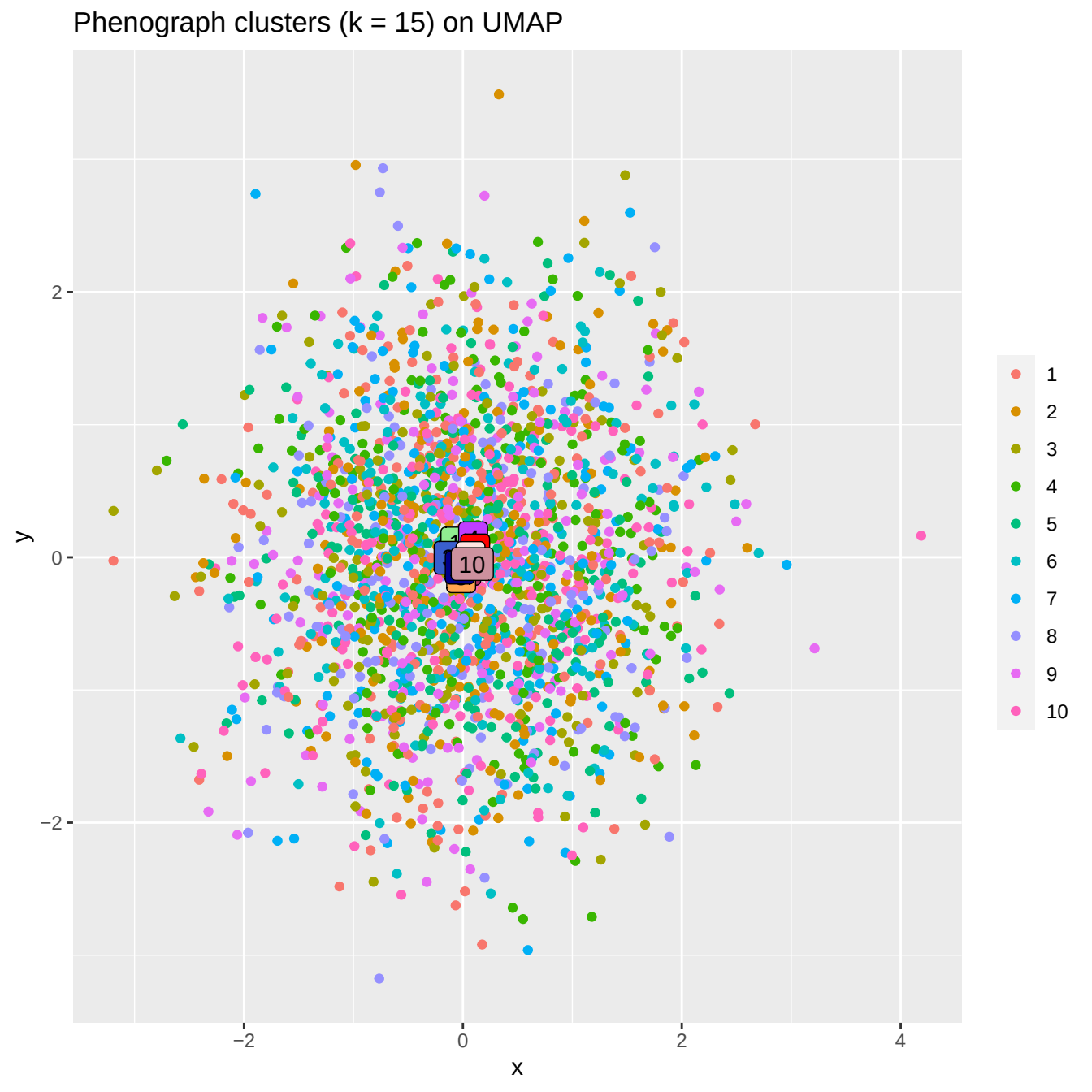
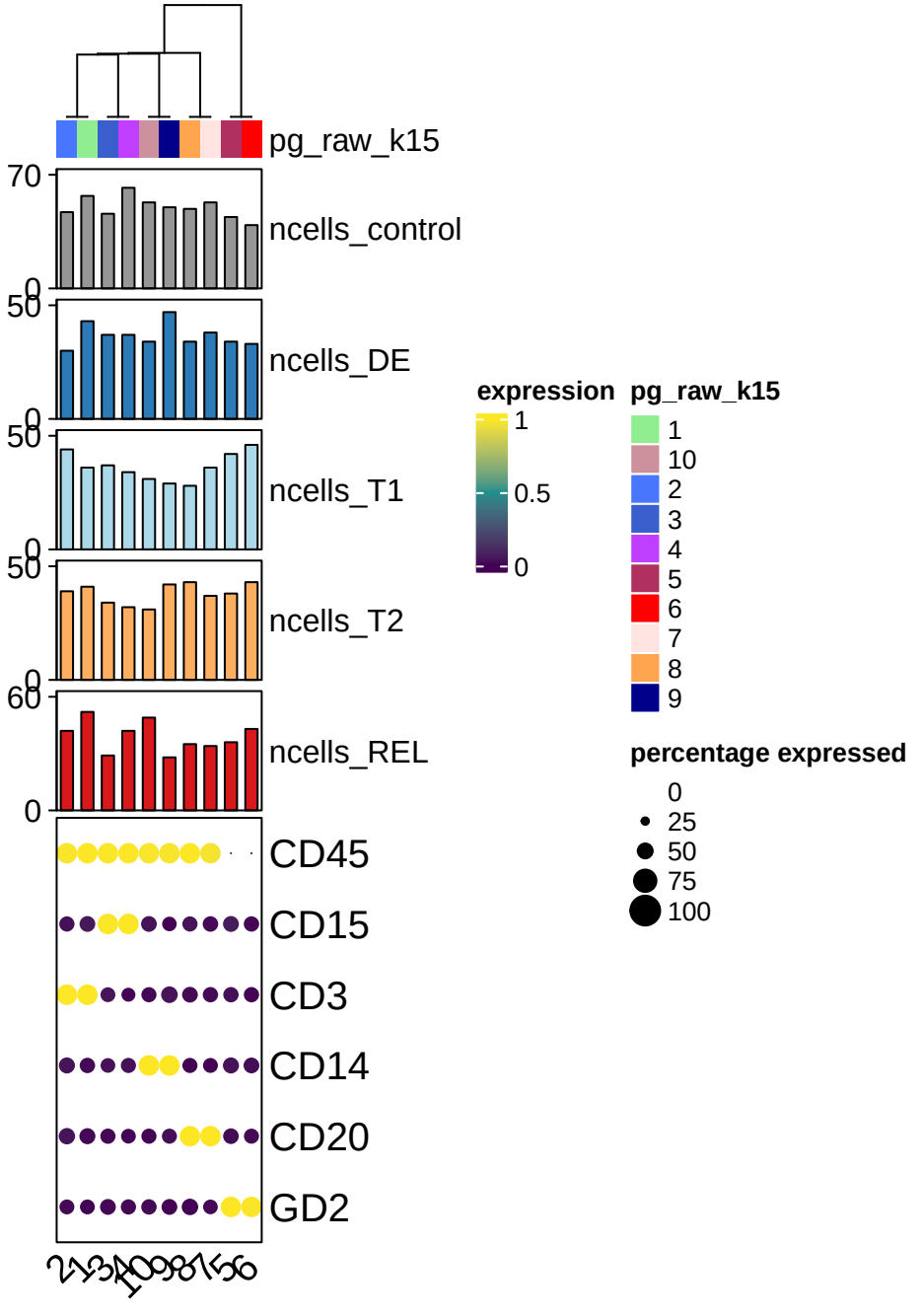
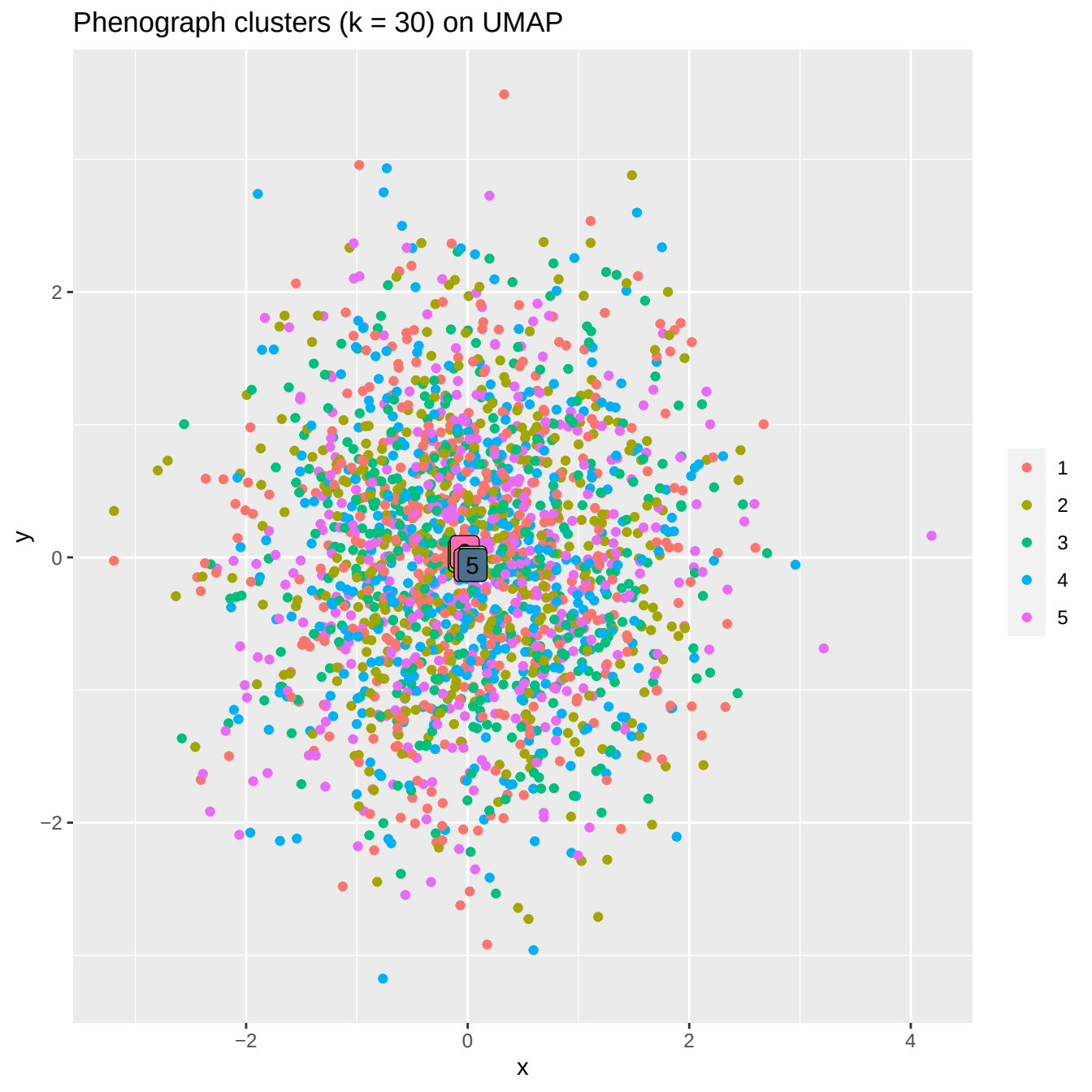
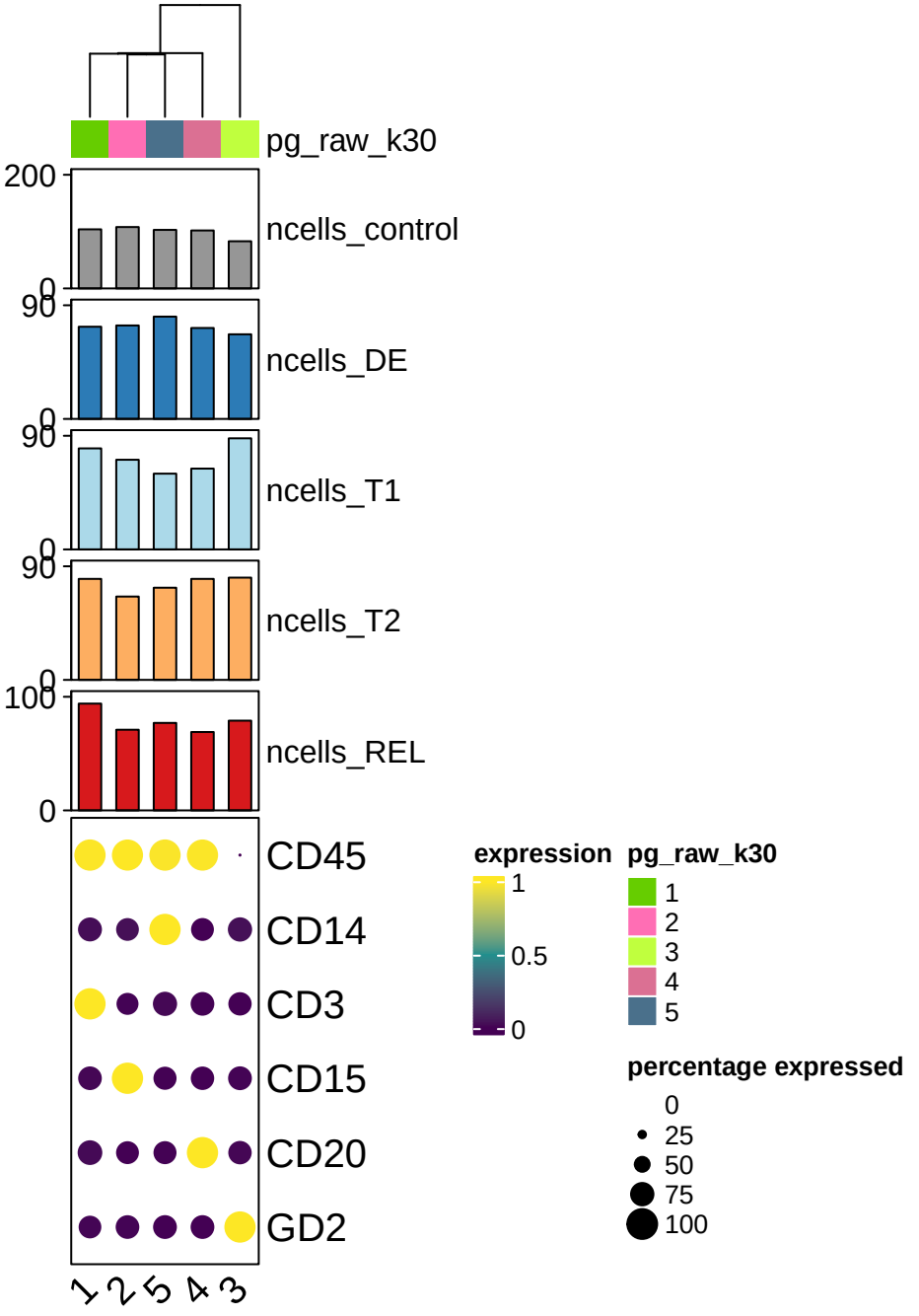
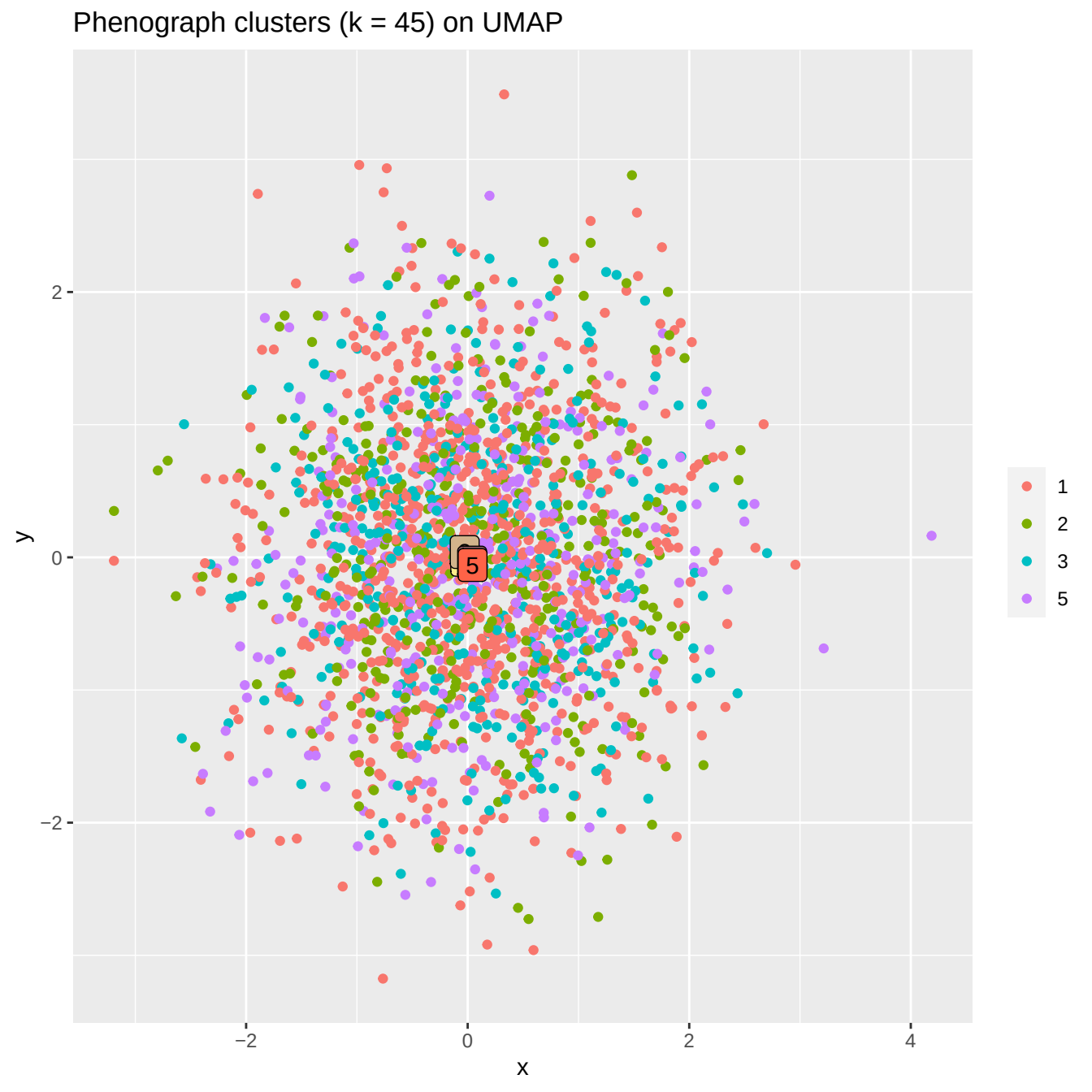
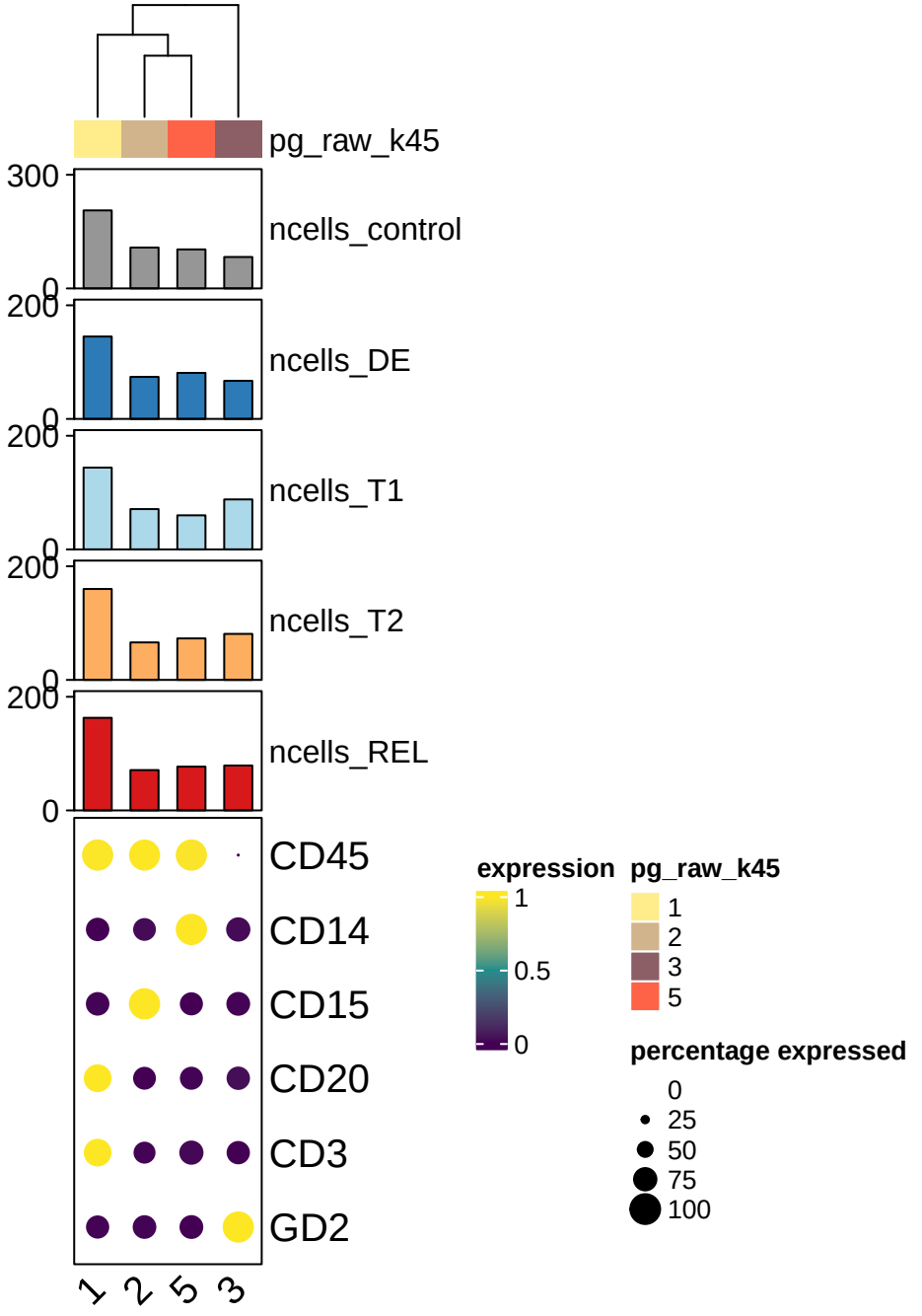

Phenograph clusters (k = 15) on UMAP
2
0
−2
y
−2
0
2
4
x
10
1
2
3
4
5
6
7
8
9
10

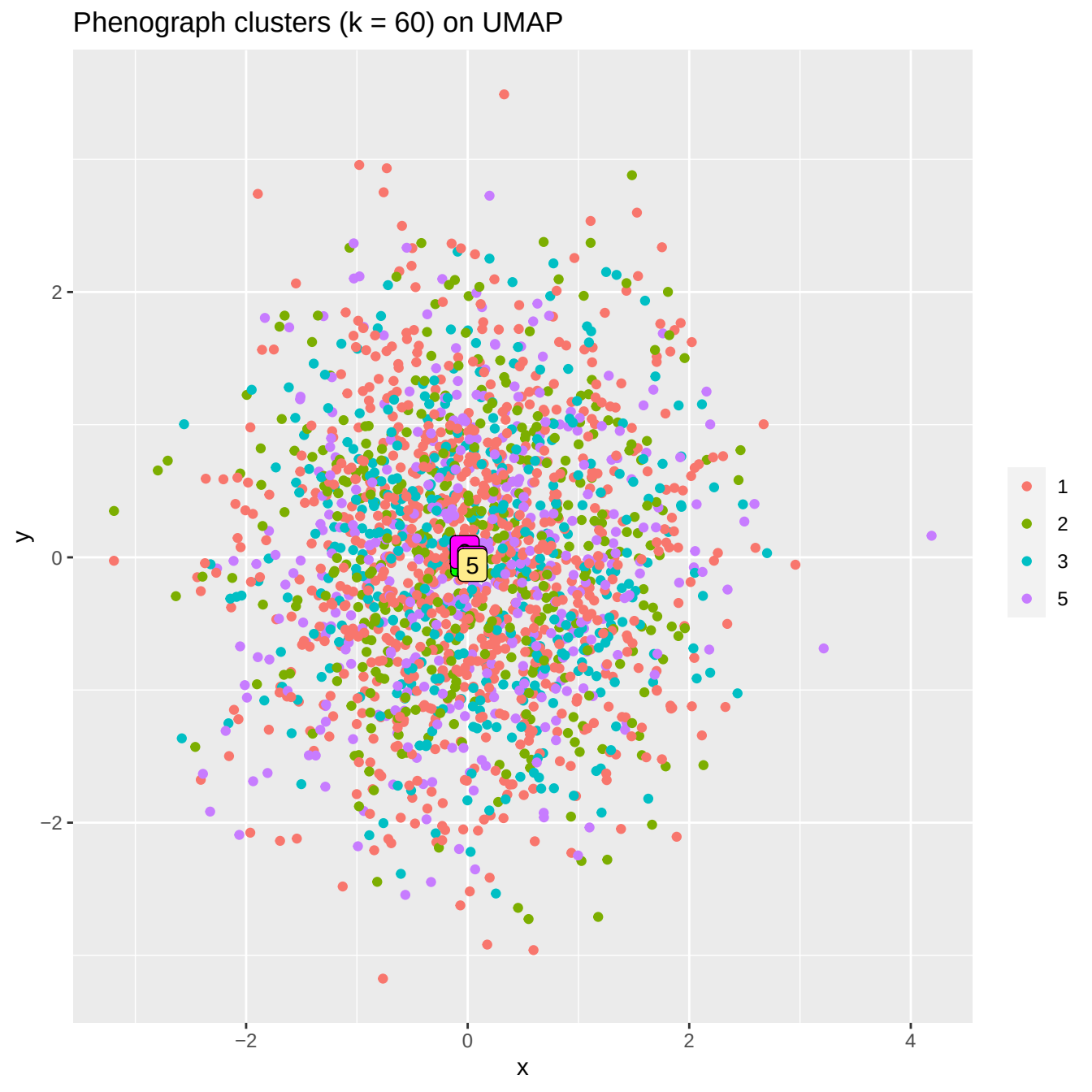

pg_raw_k15
70
0
ncells_control
50
0
ncells_DE
50
0
ncells_T1
50
0
ncells_T2
60
0
ncells_REL
CD45
CD15
CD3
CD14
CD20
GD2
expression
1
0.5
0
pg_raw_k15
1
10
2
3
4
5
6
7
8
9
percentage expressed
0
25
50
75
100

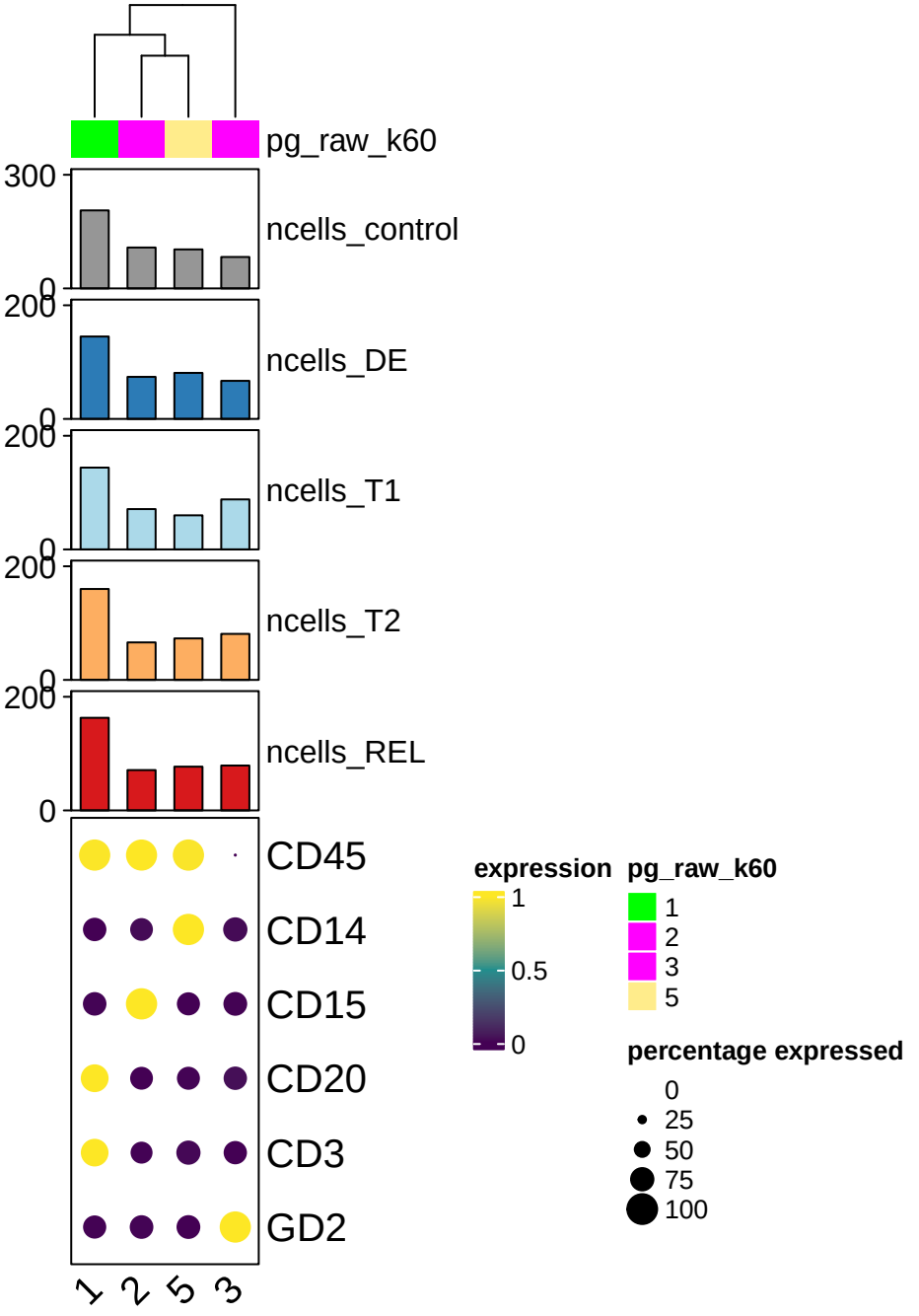

Phenograph clusters (k = 30) on UMAP
2
0
−2
y
−2
0
2
4
x
5
1
2
3
4
5

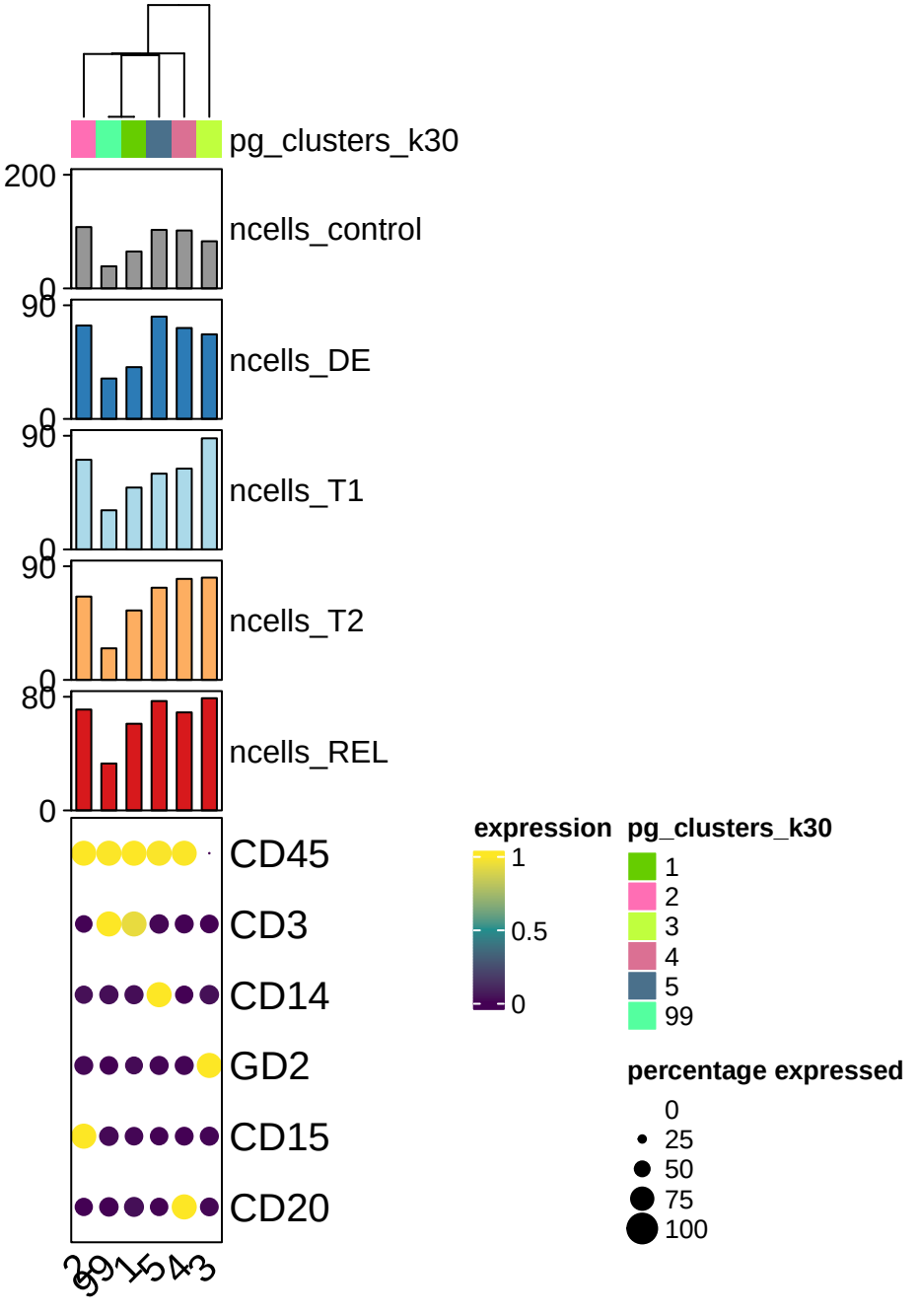

pg_raw_k30
200
0
ncells_control
90
0
ncells_DE
90
0
ncells_T1
90
0
ncells_T2
100
0
ncells_REL
CD45
CD14
CD3
CD15
CD20
GD2
1 2 5 4 3
expression
1
0.5
0
pg_raw_k30
1
2
3
4
5
percentage expressed
0
25
50
75
100

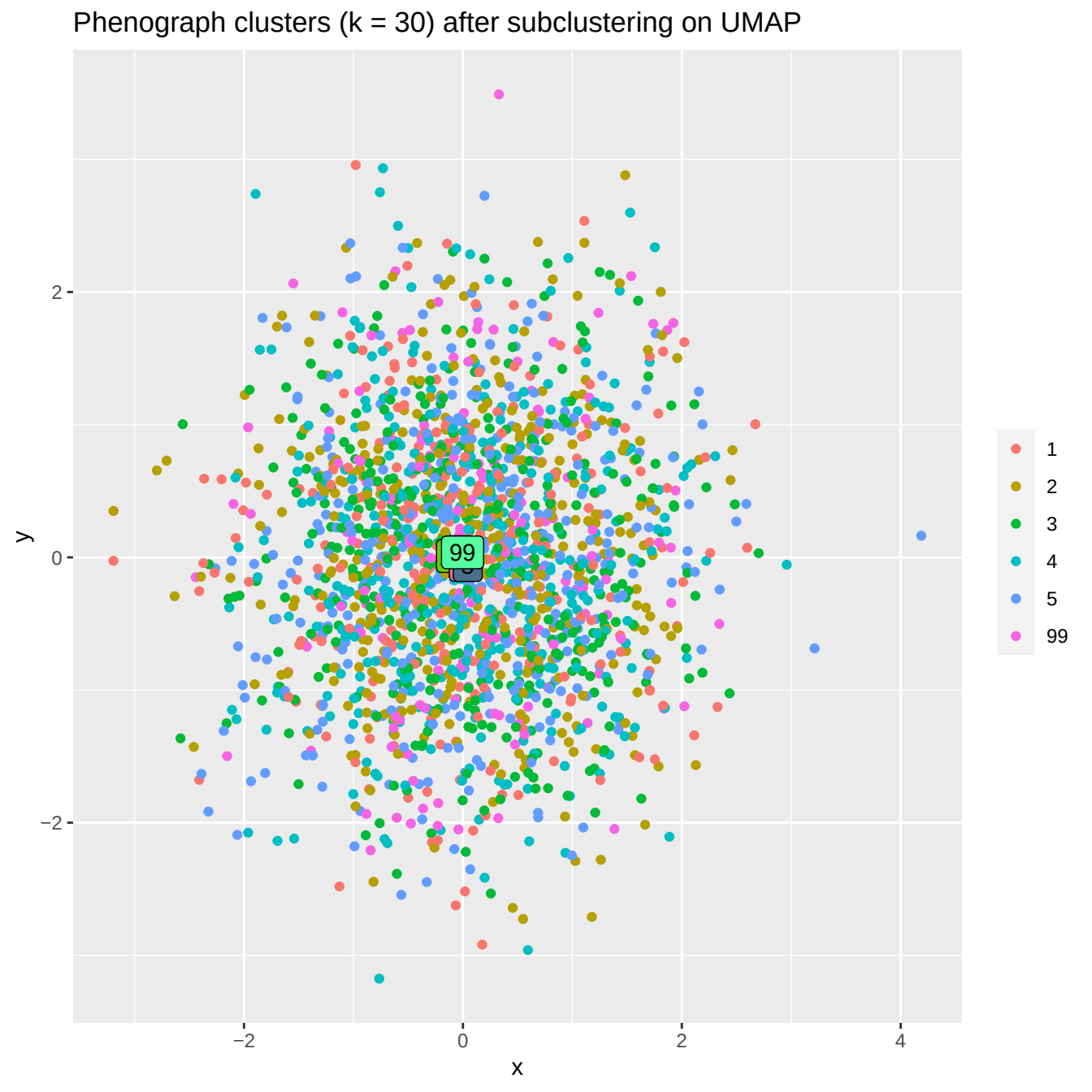

Phenograph clusters (k = 45) on UMAP
2
0
−2
y
−2
0
2
4
x
5
1
2
3
5

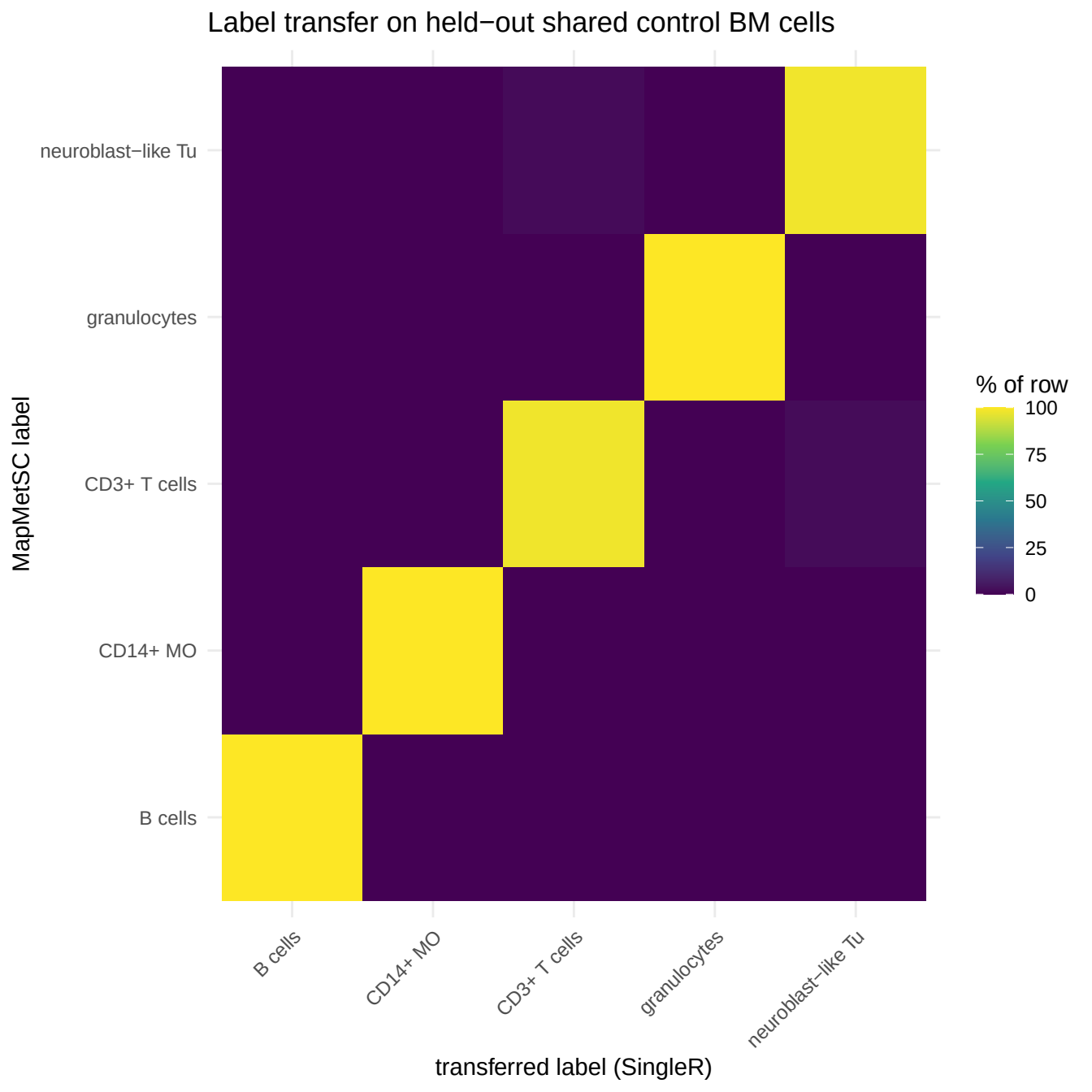

pg_raw_k45
300
0
ncells_control
200
0
ncells_DE
200
0
ncells_T1
200
0
ncells_T2
200
0
ncells_REL
CD45
CD14
CD15
CD20
CD3
GD2
1
2
5
3
expression
1
0.5
0
pg_raw_k45
1
2
3
5
percentage expressed
0
25
50
75
100

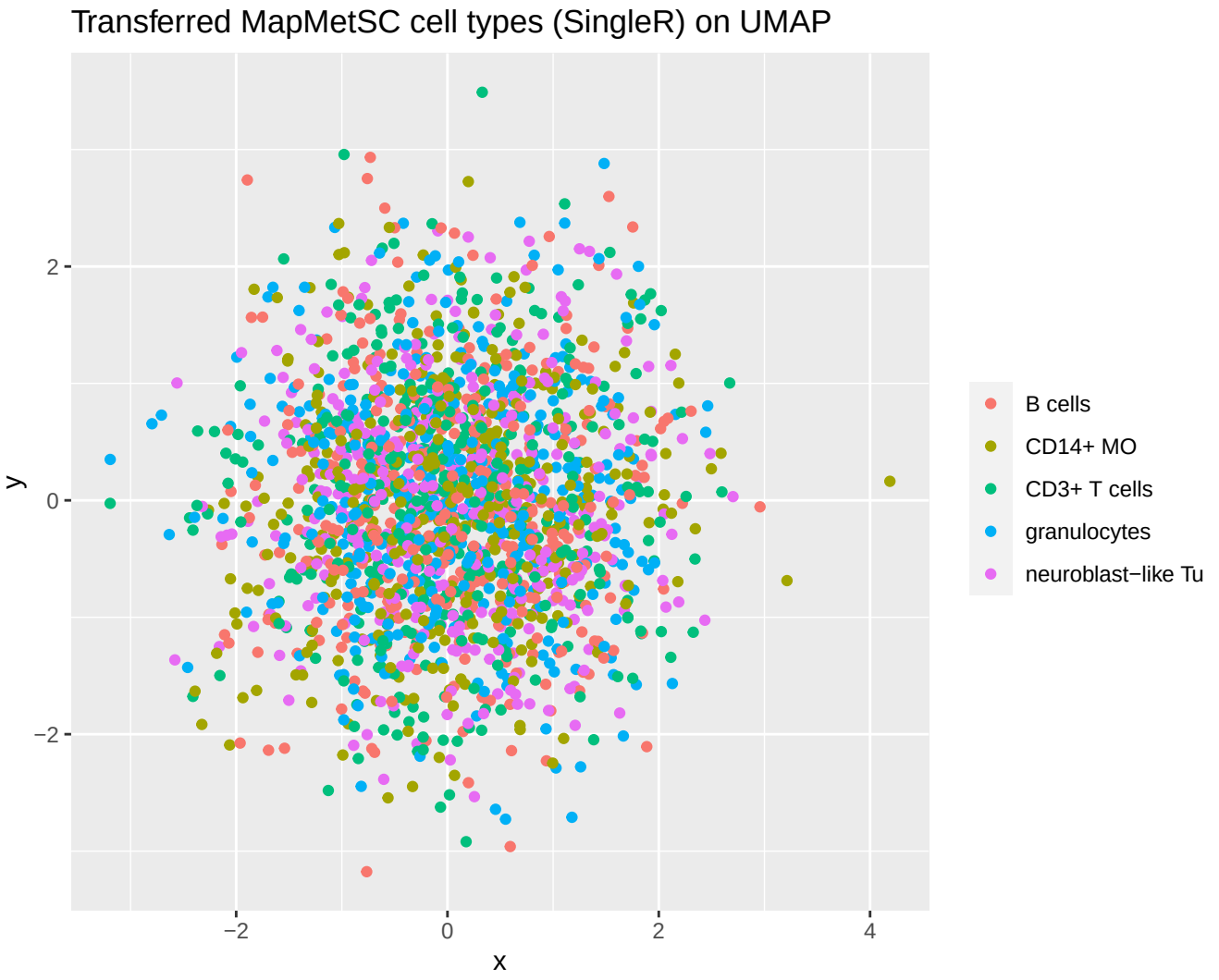

Phenograph clusters (k = 60) on UMAP
2
0
−2
y
−2
0
2
4
x
5
1
2
3
5

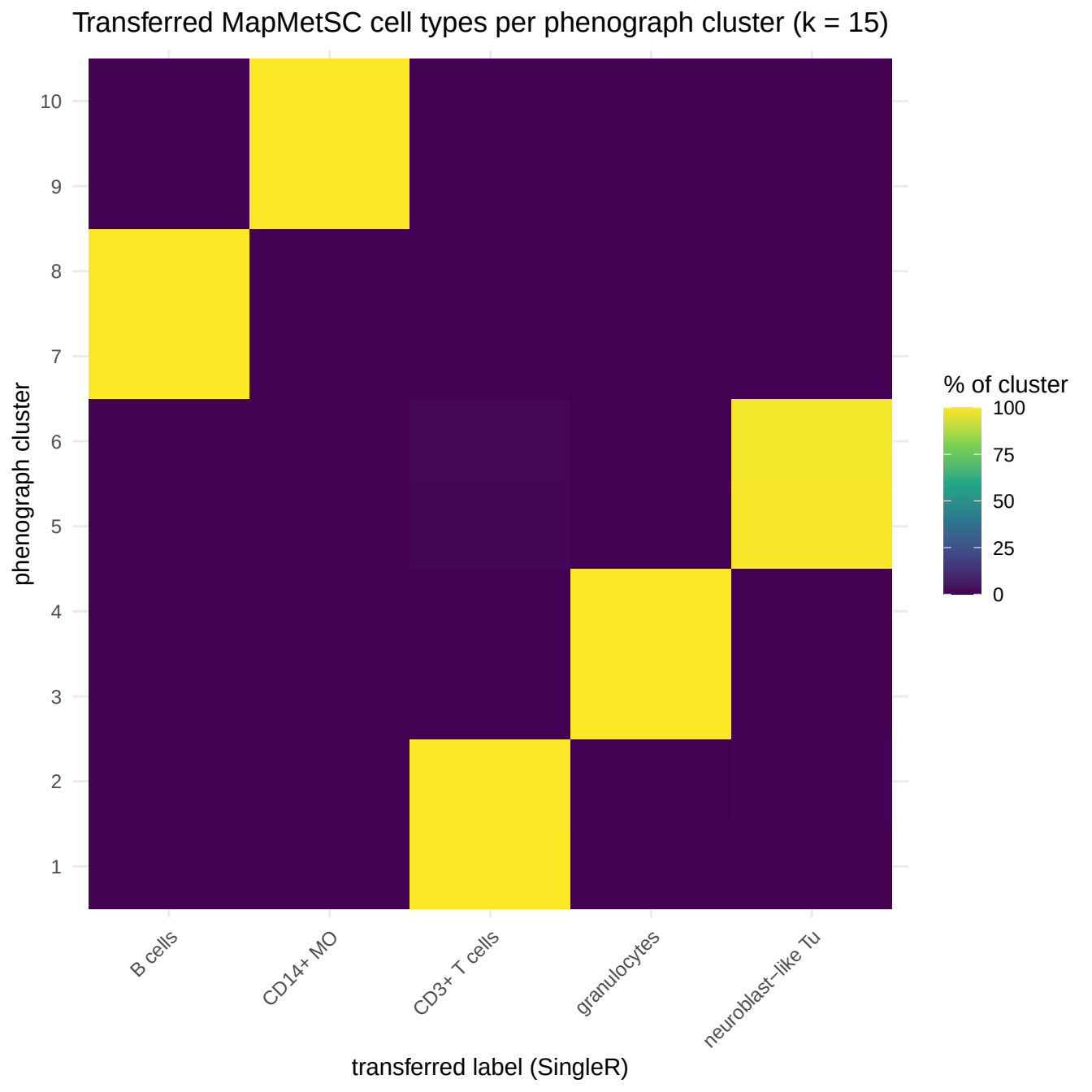

pg_raw_k60
300
0
ncells_control
200
0
ncells_DE
200
0
ncells_T1
200
0
ncells_T2
200
0
ncells_REL
CD45
CD14
CD15
CD20
CD3
GD2
1
2
5
3
expression
1
0.5
0
pg_raw_k60
1
2
3
5
percentage expressed
0
25
50
75
100

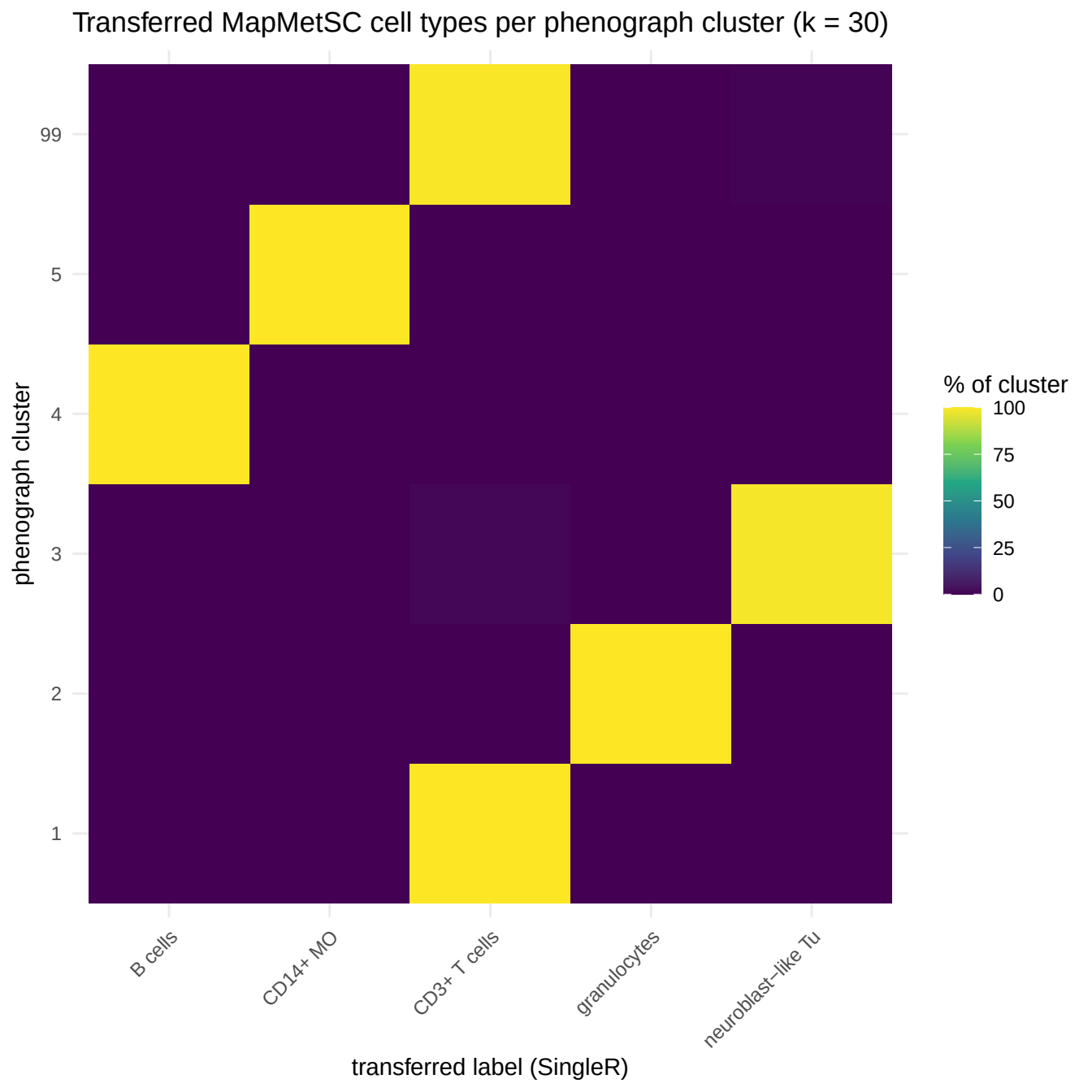

pg_clusters_k30
200
0
ncells_control
90
0
ncells_DE
90
0
ncells_T1
90
0
ncells_T2
80
0
ncells_REL
CD45
CD3
CD14
GD2
CD15
CD20
2
99
1
5
4
3
expression
1
0.5
0
pg_clusters_k30
1
2
3
4
5
99
percentage expressed
0
25
50
75
100

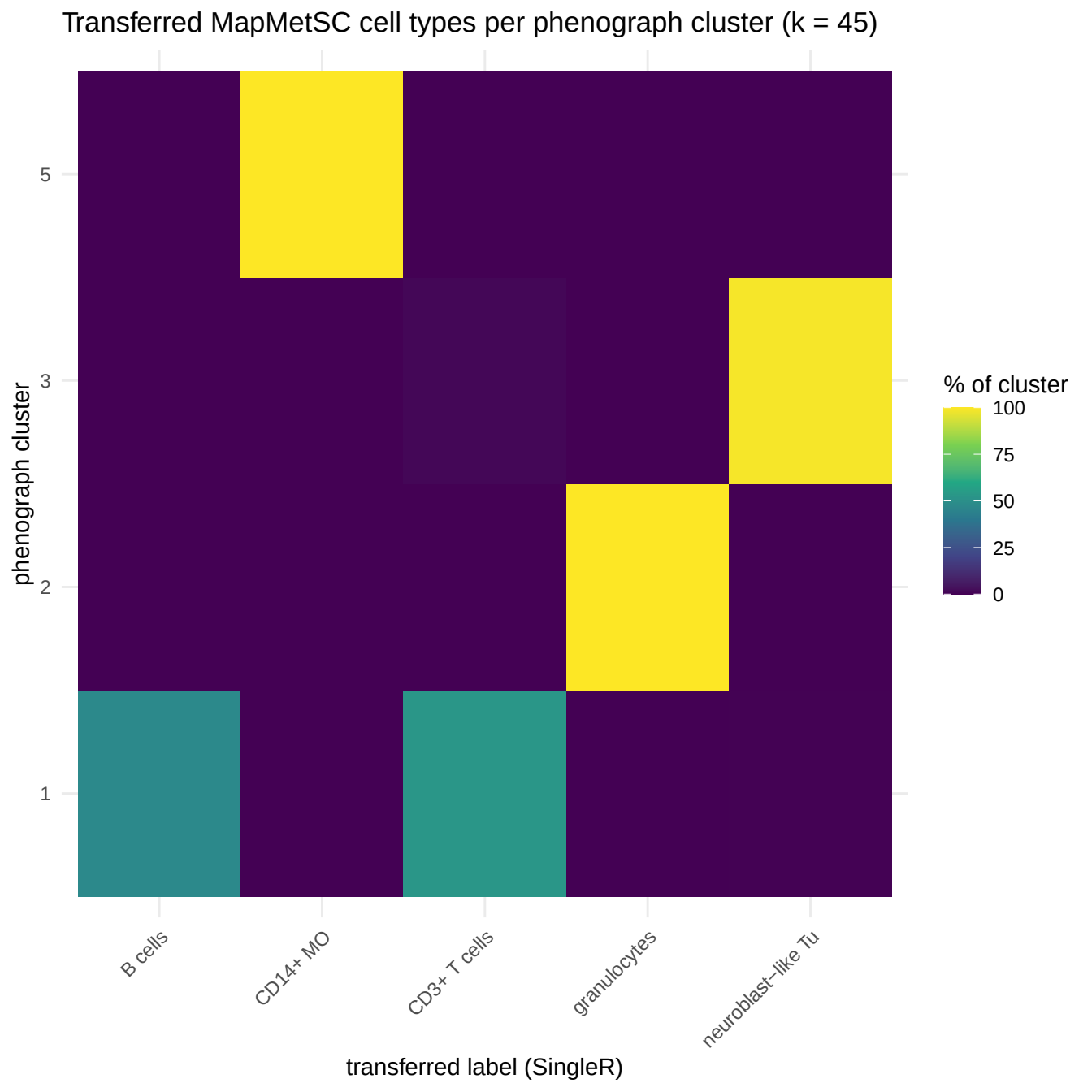

Phenograph clusters (k = 30) after subclustering on UMAP
2
0
−2
y
−2
0
2
4
x
99
1
2
3
4
5
99

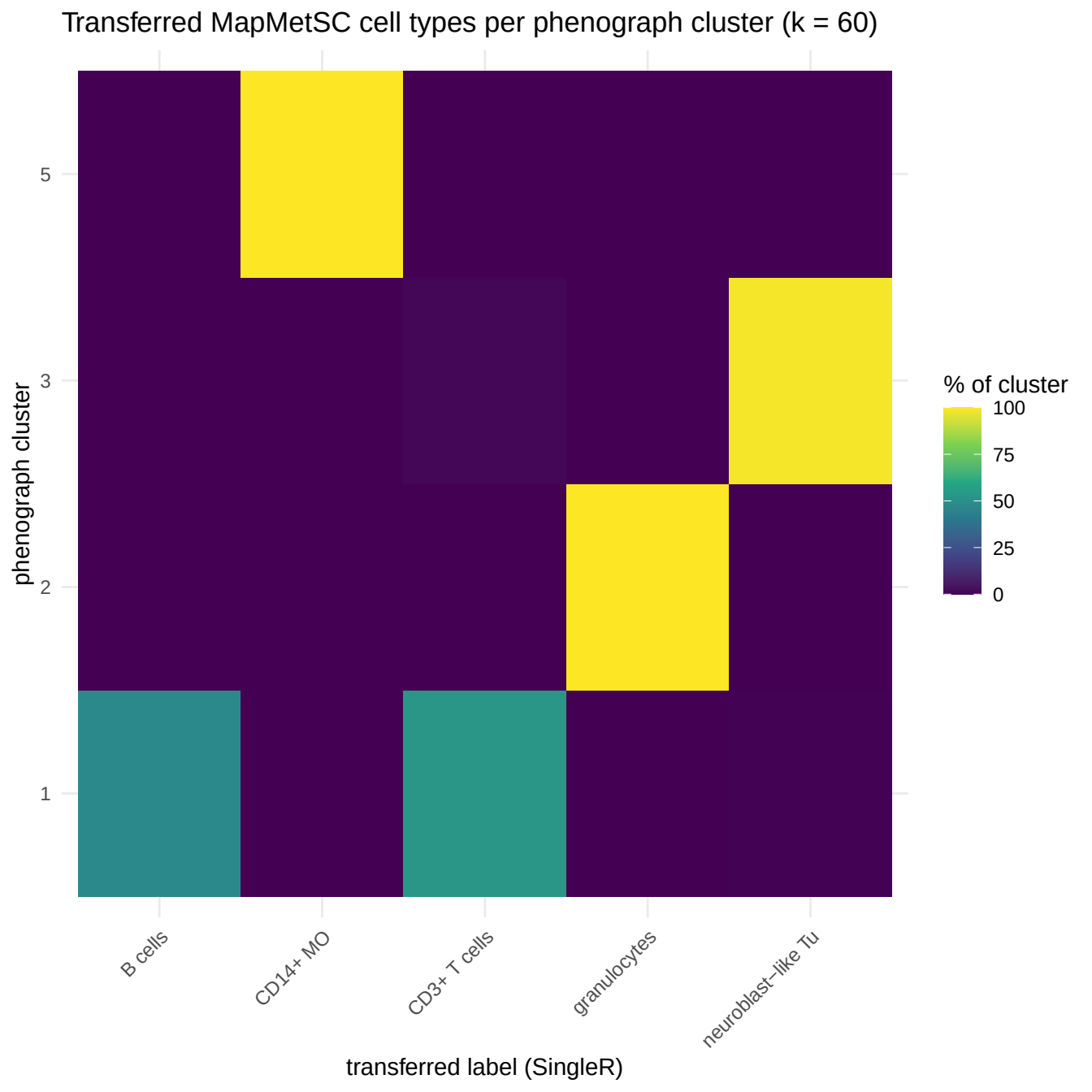

Label transfer on held−out shared control BM cells
MapMetSC label
neuroblast−like Tu
granulocytes
CD3+ T cells
CD14+ MO
B cells
transferred label (SingleR)
B cells
CD14+ MO
CD3+ T cells
granulocytes
neuroblast−like Tu
% of row
100
75
50
25
0


Transferred MapMetSC cell types (SingleR) on UMAP
y
2
0
−2
x
−2
0
2
4
B cells
CD14+ MO
CD3+ T cells
granulocytes
neuroblast−like Tu


Transferred MapMetSC cell types per phenograph cluster (k = 15)
phenograph cluster
10
9
8
7
6
5
4
3
2
1
B cells
CD14+ MO
CD3+ T cells
granulocytes
neuroblast−like Tu
transferred label (SingleR)
% of cluster
100
75
50
25
0


Transferred MapMetSC cell types per phenograph cluster (k = 30)
phenograph cluster
99
5
4
3
2
1
B cells
CD14+ MO
CD3+ T cells
granulocytes
neuroblast−like Tu
transferred label (SingleR)
% of cluster
100
75
50
25
0


Transferred MapMetSC cell types per phenograph cluster (k = 45)
phenograph cluster
5
3
2
1
B cells
CD14+ MO
CD3+ T cells
granulocytes
neuroblast−like Tu
transferred label (SingleR)
% of cluster
100
75
50
25
0


Transferred MapMetSC cell types per phenograph cluster (k = 60)
phenograph cluster
5
3
2
1
B cells
CD14+ MO
CD3+ T cells
granulocytes
neuroblast−like Tu
transferred label (SingleR)
% of cluster
100
75
50
25
0